Software Requirements Specification (SRS)

Learning Management System (LMS) Lite

1. Introduction

1.1 Purpose

The purpose of this SRS document is to define the comprehensive requirements for the Learning Management System (LMS) Lite. This system will facilitate a streamlined, secure, and efficient process for educational institutions and organizations to deliver online learning experiences, enabling instructors to create and manage courses, assessments, and student progress while providing robust tools for content delivery, student engagement, and educational analytics.

1.2 Scope

The system enables educational institutions to manage comprehensive online learning programs, course delivery, and student assessment while tracking academic progress and maintaining educational standards. It supports administrators, instructors, and students in overseeing educational journeys, managing course content, and ensuring achievement of learning objectives through role-based access control. The system will expose secured REST APIs, a web-based frontend with authentication, and comprehensive CRUD operations. Key functionalities include:

- User authentication and educational profile management system
- Role-based access control (Student, Instructor, Content Creator, Administrator)
- Complete CRUD operations for course and content management
- Course enrollment and learning path management system
- Assessment creation and grading automation
- Progress tracking and learning analytics dashboard
- Communication tools for instructor-student interaction
- Notification system for assignment deadlines and course updates
- Reporting and analytics for educational insights
- Content management with multimedia support
- Comprehensive search and filtering capabilities

1.3 Definitions, Acronyms, and Abbreviations

- **API**: Application Programming Interface
- **REST**: Representational State Transfer
- **UI**: User Interface

- **SRS**: Software Requirements Specification
- **CORS**: Cross-Origin Resource Sharing
- **JPA**: Java Persistence API
- **JWT**: JSON Web Token
- **RBAC**: Role-Based Access Control
- **CRUD**: Create, Read, Update, Delete
- **LMS**: Learning Management System
- **SCORM**: Sharable Content Object Reference Model
- **LTI**: Learning Tools Interoperability
- **FERPA**: Family Educational Rights and Privacy Act
- **GPA**: Grade Point Average
- **MOOC**: Massive Open Online Course

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- Spring Boot Security Documentation
- React.js Documentation
- Educational Technology Standards (IMS Global)
- FERPA Compliance Guidelines
- SCORM 2004 Specification
- Accessibility Standards for Educational Content (WCAG 2.1)

1.5 Overview

This document covers a high-level system overview, user roles, interface descriptions, and specific functional and non-functional requirements for the Learning Management System (LMS) Lite with enhanced security, role-based features, and comprehensive educational content management capabilities.

2. Overall Description

2.1 Product Perspective

This is a standalone web application developed using Spring Boot for backend services with Spring Security, React.js for frontend interface with authentication, and secured REST APIs for communication. The system uses JPA for database operations, JWT for authentication, and implements comprehensive role-based access control specifically designed for educational content delivery and learning management operations.

2.2 Product Functions

- User registration and secure authentication system
- Role-based dashboard interfaces for different user types
- Complete CRUD operations for course and content management
- Student enrollment and learning path workflow system
- Assessment creation and automated grading management
- Progress tracking with comprehensive learning analytics
- Communication tools for educational interaction
- Notification system for academic alerts and updates
- Reporting and analytics for educational and performance insights
- Content search and filtering with advanced educational criteria
- Learning history tracking and academic audit capabilities
- Integration with external educational tools and platforms

2.3 User Classes and Characteristics

- **System Administrator**: Full system access, user management, system configuration, and platform oversight
- **Instructor**: Course creation access, content management, student assessment, grade management, and educational functions
- **Content Creator**: Specialized content development, multimedia creation, course material design, and educational resource management
- **Teaching Assistant**: Limited instructor access, student support, assignment grading, and course assistance capabilities
- **Student**: Course enrollment, content access, assignment submission, and basic learning management
- **Guest User**: Limited access to public courses and trial functionality

2.4 Operating Environment

- **Server**: Spring Boot 3.x with Spring Security (Java 17+)
- **Database**: JPA-compatible database with educational management tables
- **Frontend**: React.js with authentication components (port 8081)
- **Backend**: Spring Boot with security (port 8080)
- **Authentication**: JWT-based authentication system
- **Testing**: JUnit for backend, React Testing Library for frontend testing

2.5 Design and Implementation Constraints

- RESTful API design with security features
- JWT-based authentication required for all protected operations
- Role-based access control implementation mandatory
- Comprehensive input validation and sanitization
- Educational data storage and academic record validation requirements
- FERPA compliance with student data protection standards
- Real-time progress tracking mandatory
- Mobile-responsive design for cross-device learning accessibility
- Integration with educational standards and external learning tools

2.6 User Documentation

- User registration and course management guides
- Role-specific user manuals
- Administrator system management guide
- Instructor course creation and management guides
- Student learning and assessment guides
- Privacy policy and educational data documentation

2.7 Assumptions and Dependencies

- Users have secure internet access and modern web browsers
- Integration with educational technology systems
- Email service integration for notifications
- Backup and recovery service availability
- Future integration with library systems and educational databases

3. Specific Requirements

3.1 External Interfaces

3.1.1 User Interfaces

- **Landing/Home Page**: Educational platform overview with user registration and login options
- **User Registration**: Account creation with validation and verification
- **Login/Authentication**: Secure login with role-based dashboard redirection
- **Learning Dashboard**: Course management, progress tracking, and analytics
- **Admin Panel**: User management, system configuration, and platform oversight
- **Course Management**: Course creation, content uploading, and management interface

- **Assessment Center**: Quiz and assignment creation, grading interface
- **Student Portal**: Course access, assignment submission, and progress viewing

3.1.2 Hardware Interfaces

- Document scanner integration for assignment submission
- Webcam integration for proctored assessments
- Audio/video recording devices for multimedia content

3.1.3 Software Interfaces

- Secured REST API with JWT authentication
- Educational platform and LTI integration
- Email service integration for notifications
- Cloud storage integration for content management
- Video streaming service integration

3.1.4 Communication Interfaces

- HTTPS protocols with SSL/TLS encryption
- JSON data format with security headers
- JWT token-based authentication
- Real-time notification delivery system
- WebSocket connections for live interactions

3.2 Functional Requirements

User Authentication and Account Management

FR1: Comprehensive User Registration and Authentication System

Description: The system shall provide a comprehensive user registration mechanism that captures essential user information, validates credentials, and ensures compliance with educational platform access standards. This functionality serves as the foundation for establishing trusted user relationships and maintaining access control throughout the educational platform.

Detailed Requirements:

- The registration form shall capture complete user information including full name, email address, educational institution, role type, department affiliation, and contact information
- User verification shall require email confirmation and institutional validation where applicable
- User profile creation shall include personal information, educational background, role preferences, and notification settings

- Educational platform compliance shall validate user eligibility, institutional affiliation, and policy acknowledgment
- Account activation workflow shall include automated verification followed by optional manual review for instructor approval
- Multi-role support shall handle student registration, instructor accounts, and administrative accounts with appropriate verification levels
- Password security shall enforce strong password policies with complexity requirements and periodic updates
- Account recovery shall provide secure password reset mechanisms with multi-factor verification
- Terms and conditions acceptance shall include privacy policies, academic integrity agreements, data usage terms, and platform policies
- User profile management shall allow updates to personal information, preferences, and communication settings

FR2: Secure Authentication and Session Management System

Description: The system shall implement a robust authentication mechanism designed for educational environments, incorporating security measures appropriate for academic settings while providing seamless access to learning management functionalities and maintaining high security standards for sensitive educational data.

Detailed Requirements:

- Multi-credential login shall accept user ID, email address, or student/employee ID as valid login credentials for user convenience
- JWT-based authentication shall provide secure token management with role-specific access tokens, refresh token rotation, and education-appropriate session durations
- Academic-optimized security shall include optional two-factor authentication, trusted device management, and institutional IP whitelisting for enhanced security
- Session management shall maintain user login status across educational operations while implementing automatic logout after extended inactivity periods
- Device recognition shall track user devices and require additional verification for new or suspicious login attempts
- Progressive security measures shall implement escalating lockout periods for failed attempts with academic continuity considerations
- Single sign-on integration shall support institutional authentication systems and educational directory services
- Security logging shall maintain detailed authentication records for security monitoring and compliance auditing

- Account lockout protection shall prevent brute force attacks while maintaining legitimate user access
- Integration with educational systems shall support connection with existing student information systems and institutional authentication platforms

FR3: Role-Based Access Control for Educational Management Operations

Description: The system shall implement a specialized role-based access control system tailored for educational management operations, ensuring appropriate access levels for different user types while maintaining operational efficiency, academic integrity, and data security throughout the platform.

Detailed Requirements:

- Student role permissions shall include course enrollment, content access, assignment submission, grade viewing, and personal profile updates with restrictions on administrative functions
- Instructor role shall have elevated permissions for course creation, content management, student assessment, grade assignment, and educational analytics
- Content Creator role shall have specialized permissions for multimedia content development, course material creation, and educational resource management
- Teaching Assistant role shall have limited permissions for student support, assignment grading, discussion moderation, and course assistance within assigned courses
- System Administrator role shall have complete access including user management, system configuration, academic policies, and full operational oversight
- Guest user access shall be limited to public course browsing, demo content, and account creation without access to enrolled course content
- Hierarchical permission structure shall support department-based access control with configurable permissions for different educational functions
- Temporal access controls shall implement academic term restrictions, maintenance windows, and emergency access procedures
- Function-based permissions shall control access to specific operations like grade modification, bulk updates, and sensitive academic reporting
- Data access controls shall ensure users can only access appropriate data based on their role and relationship to the educational content
- Audit trail integration shall log all role-based access events for security monitoring and compliance reporting

Course and Content Management

FR4: Comprehensive Course Creation and Management System

Description: The system shall provide a sophisticated course management interface that allows instructors to create detailed educational content while ensuring course quality, validation compliance,

and comprehensive course documentation throughout the academic lifecycle.

Detailed Requirements:

- Course creation form shall capture comprehensive details including course title, description, learning objectives, prerequisites, credit hours, and duration
- Data validation shall enforce required field completion, course code format validation, realistic schedule constraints, and academic policy compliance
- Course structure management shall support modules, lessons, topics, and learning units with hierarchical organization
- Content type validation shall ensure appropriate formats for lectures, readings, videos, assignments, and assessments
- Course scheduling workflow shall implement start dates, end dates, class schedules, and academic calendar integration
- Course difficulty assessment shall provide beginner, intermediate, and advanced level indicators based on content complexity
- Course dependency management shall handle prerequisite courses, sequential learning paths, and interconnected course requirements
- Course modification workflow shall implement course updates, content revisions, and schedule adjustments
- Course analytics shall track enrollment rates, completion rates, and course effectiveness metrics
- Content versioning shall maintain course history, content revisions, and update tracking
- Course template system shall provide pre-designed course structures for common academic subjects

FR5: Learning Content Management and Delivery System

Description: The system shall provide comprehensive content management capabilities that handle educational material storage, multimedia content delivery, and learning resource organization while ensuring content quality and accessibility standards.

Detailed Requirements:

- Content upload workflow shall support various file formats including documents, videos, audio files, presentations, and interactive content
- Multimedia content management shall handle video streaming, audio playback, interactive simulations, and rich media presentations
- Content organization shall provide folder structures, tagging systems, categorization, and search functionality
- Version control shall track content updates, revision history, and collaborative editing capabilities

- Content accessibility shall ensure compliance with accessibility standards, alternative text for images, and screen reader compatibility
- Content security shall implement access controls, download restrictions, and content protection measures
- Learning path creation shall enable sequential content delivery, adaptive learning paths, and personalized learning experiences
- Content analytics shall monitor content engagement, viewing time, completion rates, and learning effectiveness
- Resource library shall provide shared content repository, reusable learning objects, and collaborative content development
- Integration capabilities shall connect with external content providers, library systems, and educational resource databases

Assessment and Evaluation Management

FR6: Comprehensive Assessment Creation and Management System

Description: The system shall implement comprehensive assessment tools that handle quiz creation, assignment management, grading workflows, and academic evaluation while maintaining assessment integrity and supporting diverse evaluation methods.

Detailed Requirements:

- Assessment creation shall support multiple question types including multiple choice, true/false, essay, short answer, and multimedia questions
- Quiz management shall provide timed assessments, attempt limitations, randomized question pools, and anti-cheating measures
- Assignment workflow shall handle submission requirements, due dates, late submission policies, and plagiarism detection integration
- Grading system shall support rubric-based evaluation, automated grading for objective questions, and manual grading for subjective content
- Grade management shall calculate course grades, weighted averages, grade curves, and academic performance analytics
- Assessment analytics shall analyze question effectiveness, difficulty levels, discrimination indices, and learning outcome achievement
- Feedback system shall enable detailed instructor feedback, peer review capabilities, and self-assessment tools
- Academic integrity shall implement proctoring options, browser lockdown, and suspicious activity detection

- Grade book management shall maintain comprehensive grade records, attendance tracking, and academic progress monitoring
- Assessment scheduling shall coordinate exam periods, assignment deadlines, and academic calendar integration

Student Progress and Analytics

FR7: Comprehensive Learning Progress Tracking System

Description: The system shall provide comprehensive progress tracking capabilities that monitor student advancement, analyze learning patterns, and generate educational insights through detailed learning analytics and performance monitoring tools.

Detailed Requirements:

- Progress monitoring shall track course completion, module advancement, assignment submissions, and learning milestone achievement
- Learning analytics shall analyze study patterns, engagement levels, time spent on content, and learning effectiveness indicators
- Performance visualization shall provide progress charts, grade trends, competency mapping, and achievement dashboards
- Predictive analytics shall identify at-risk students, intervention opportunities, and success probability indicators
- Competency tracking shall monitor skill development, learning objective achievement, and knowledge gap identification
- Engagement metrics shall measure participation levels, discussion contributions, content interaction, and platform usage patterns
- Academic reporting shall generate transcript-like records, progress reports, and academic achievement summaries
- Comparative analysis shall benchmark student performance against class averages, historical data, and learning standards
- Intervention alerts shall notify instructors of student difficulties, missed deadlines, and performance concerns
- Learning pathway optimization shall recommend personalized learning routes based on individual progress and preferences

Communication and Collaboration Tools

FR8: Educational Communication and Collaboration System

Description: The system shall implement comprehensive communication tools that enable instructor-student interaction, peer collaboration, and academic community building while maintaining professional boundaries and educational focus.

Detailed Requirements:

- Discussion forums shall provide threaded conversations, topic-based discussions, Q&A sections, and moderated academic discourse
- Messaging system shall facilitate direct instructor-student communication, group messaging, and academic support conversations
- Announcement system shall deliver course updates, assignment notifications, institutional announcements, and emergency communications
- Video conferencing integration shall support virtual classrooms, office hours, group meetings, and collaborative sessions
- Collaborative tools shall enable group projects, peer review activities, shared workspaces, and team-based learning
- Office hours management shall schedule instructor availability, appointment booking, and virtual meeting coordination
- Academic support channels shall provide tutoring requests, technical support, and academic advising communications
- Content sharing shall enable resource sharing, study group formation, and collaborative content creation
- Communication analytics shall track engagement levels, response times, and communication effectiveness
- Professional communication standards shall ensure appropriate academic discourse and professional interaction guidelines

Administrative and Reporting Functions

FR9: Comprehensive Administrative Management System

Description: The system shall provide extensive administrative capabilities that handle institutional management, user administration, academic policy enforcement, and operational oversight for effective educational platform governance.

Detailed Requirements:

- User account lifecycle management shall handle enrollment, registration, role assignments, and account deactivation with appropriate approval workflows
- Course catalog management shall organize course offerings, academic programs, prerequisite tracking, and curriculum planning

- Academic calendar integration shall manage semester schedules, holiday periods, exam schedules, and important academic dates
- Enrollment management shall handle course registration, waitlist management, capacity controls, and enrollment reporting
- Grade processing shall automate transcript generation, GPA calculations, academic standing determination, and degree audit support
- Institutional reporting shall provide enrollment statistics, academic performance metrics, and compliance reporting
- System configuration shall allow customization of academic policies, grading scales, institutional branding, and platform settings
- Data backup and archiving shall ensure academic record preservation, compliance with retention policies, and disaster recovery
- Integration management shall oversee connections with student information systems, library systems, and external educational tools
- Compliance monitoring shall ensure adherence to FERPA requirements, accessibility standards, and institutional policies

FR10: Advanced Educational Analytics and Intelligence System

Description: The system shall implement comprehensive educational intelligence capabilities that provide advanced insights, institutional analytics, and decision support tools for academic improvement, curriculum optimization, and educational effectiveness enhancement.

Detailed Requirements:

- Institutional analytics shall analyze enrollment trends, course effectiveness, instructor performance, and student success metrics
- Curriculum analytics shall evaluate course sequences, learning outcome achievement, prerequisite effectiveness, and program completion rates
- Learning effectiveness measurement shall assess teaching methods, content engagement, assessment validity, and educational goal achievement
- Resource utilization analysis shall monitor platform usage, content effectiveness, feature adoption, and system performance metrics
- Predictive modeling shall forecast enrollment trends, student success probability, and resource planning requirements
- Benchmark analysis shall compare institutional performance against peer institutions, industry standards, and best practices
- Quality assurance metrics shall track course quality indicators, instructor effectiveness, and student satisfaction measures

- Data-driven recommendations shall provide insights for curriculum improvement, resource allocation, and strategic planning
- Compliance reporting shall generate required institutional reports, accreditation documentation, and regulatory submissions
- Advanced visualization shall provide sophisticated dashboards, institutional scorecards, and interactive analytical displays

3.3 Performance Requirements

- **Authentication**: Login/logout operations < 1 second for 95% of requests
- **Course Management**: Course creation and updates < 2 seconds for 90% of requests
- **Content Delivery**: Video streaming and content loading < 3 seconds for standard content
- **Assessment Operations**: Quiz submissions and grading < 1.5 seconds for typical assessments
- **Concurrent Users**: Support 2,000+ concurrent users without performance degradation
- **Database**: Handle 50,000+ users and 10,000+ courses with efficient querying
- **Mobile Performance**: Mobile application load time < 3 seconds on 3G networks
- **Real-time Updates**: Live notifications and updates with < 1 second latency

3.4 Logical Database Requirements

Core Tables:

- **Users**: id, username, email, passwordHash, role, studentId, institutionId, department, createdDate, lastLogin, isActive
- **Courses**: id, courseCode, title, description, instructorId, credits, startDate, endDate, status, maxEnrollment
- **Enrollments**: id, courseId, studentId, enrollmentDate, status, grade, completionDate
- **CourseContent**: id, courseId, title, contentType, filePath, orderIndex, isPublished, createdDate
- **Assessments**: id, courseId, title, description, type, totalPoints, dueDate, timeLimit, attempts
- **AssessmentSubmissions**: id, assessmentId, studentId, submissionDate, score, attempt, feedback
- **Discussions**: id, courseId, title, description, createdBy, createdDate, isLocked
- **DiscussionPosts**: id, discussionId, userId, content, parentPostId, createdDate
- **Grades**: id, studentId, courseId, assessmentId, points, maxPoints, gradedBy, gradedDate
- **Notifications**: id, userId, message, type, isRead, createdDate, courseId

3.5 Design Constraints

- Must implement Spring Security with JWT authentication
- Frontend must use React with protected routes and mobile-responsive design

- Database operations must use JPA with transaction management
- All academic operations must be logged for audit and compliance purposes
- Student data validation and FERPA compliance required
- Real-time progress tracking mandatory
- Integration with educational standards and external systems required
- Multi-role access control implementation required
- Accessibility compliance for educational content

3.6 Software System Attributes

3.6.1 Reliability

- 99.9% uptime during academic hours
- Graceful error handling with user-friendly messages
- Data integrity validation at all levels
- Automatic backup for critical academic data
- Comprehensive backup and disaster recovery procedures

3.6.2 Availability

- 24/7 availability with scheduled maintenance windows
- Load balancing support for high availability
- Geographic redundancy for disaster recovery
- Real-time system health monitoring
- Scalable architecture for growing educational needs

3.6.3 Security

- **Authentication**: JWT-based with educational-grade security
- **Authorization**: Role-based access control for all operations
- **Data Protection**: Encryption at rest and in transit for sensitive academic data
- **Input Validation**: Comprehensive sanitization and validation
- **Academic Data Security**: Protection of student educational records
- **Audit Trail**: Complete logging of security-relevant events
- **Compliance**: FERPA compliance and educational data protection regulation adherence

3.6.4 Maintainability

- Modular architecture with clear separation of concerns

- Comprehensive unit and integration testing
- Documentation and integration guides
- Version control with feature branching
- Automated deployment and monitoring systems

3.6.5 Portability

- Docker containerization support for easy deployment
- Database-agnostic design through JPA
- Cloud deployment ready (AWS, Azure, GCP)
- Environment-specific configuration management
- Cross-platform mobile web application compatibility

3.7 Other Requirements

- **Testing Coverage**: Minimum 85% code coverage for both backend and frontend
- **Security Testing**: Regular penetration testing and vulnerability assessment
- **Performance Testing**: Load testing for expected user and transaction volumes
- **Documentation**: Complete documentation with integration examples
- **Compliance**: FERPA compliance and educational data protection law adherence
- **Accessibility**: Web Content Accessibility Guidelines (WCAG) 2.1 compliance

4. Appendices

Appendix A: Role-Based Permission Matrix

Functionality	Guest	Student	Teaching Assistant	Instructor	Content Creator	Admin
View Public Courses	✓	✓	✓	✓	✓	✓
Create Account	✓	✓	-	-	-	-
Enroll in Courses	-	✓	-	-	-	✓
Access Course Content	-	✓	✓	✓	✓	✓
Submit Assignments	-	✓	-	-	-	-
View Grades	-	✓ (Own)	-	✓	-	✓
Create Courses	-	-	-	✓	✓	✓
Grade Assignments	-	-	✓ (Assigned)	✓	-	✓
Manage Content	-	-	✓ (Limited)	✓	✓	✓
Create Assessments	-	-	-	✓	✓	✓
View Analytics	-	-	✓ (Limited)	✓	✓	✓
Manage Users	-	-	-	-	-	✓
System Configuration	-	-	-	-	-	✓

Appendix B: Database Schema Details

Users Table:

- id (PRIMARY KEY, AUTO_INCREMENT)
- username (VARCHAR(50), UNIQUE, NOT NULL)
- email (VARCHAR(100), UNIQUE, NOT NULL)
- password_hash (VARCHAR(255), NOT NULL)
- role (ENUM: STUDENT, TEACHING_ASSISTANT, INSTRUCTOR, CONTENT_CREATOR, ADMIN)
- student_id (VARCHAR(20), UNIQUE)
- institution_id (VARCHAR(20))
- department (VARCHAR(100))
- first_name (VARCHAR(50), NOT NULL)
- last_name (VARCHAR(50), NOT NULL)
- created_date (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)
- last_login (TIMESTAMP)
- is_active (BOOLEAN, DEFAULT TRUE)

Courses Table:

- id (PRIMARY KEY, AUTO_INCREMENT)
- course_code (VARCHAR(20), UNIQUE, NOT NULL)

- title (VARCHAR(200), NOT NULL)
- description (TEXT)
- instructor_id (FOREIGN KEY → Users.id)
- credits (INT, DEFAULT 3)
- start_date (DATE, NOT NULL)
- end_date (DATE, NOT NULL)
- status (ENUM: DRAFT, PUBLISHED, ARCHIVED, CANCELLED)
- max_enrollment (INT, DEFAULT 30)
- created_date (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)

Appendix C: Validation Rules Summary

Field	Validation Rules	Error Messages
Course Code	Required, 6-20 characters, alphanumeric with hyphens	"Course code is required" / "Invalid course code format"
Course Title	Required, 5-200 characters	"Course title is required"
Email	Required, valid format, unique	"Please enter a valid email address"
Student ID	Required if student, unique, 3-20 characters	"Valid student ID is required"
Assignment Score	Required, 0-100 range, numeric	"Score must be between 0-100"
Due Date	Required, future date	"Due date must be in the future"
Course Credits	Required, 1-6 range, integer	"Credits must be between 1-6"

Appendix D: Security Implementation Details

JWT Configuration:

- Token Expiration: 8 hours for students, 12 hours for instructors, 6 hours for admins
- Refresh Token: 7 days
- Algorithm: HS256
- Secret Key: Environment variable (minimum 256 bits)
- Token Blacklisting: Implemented for logout and security breaches

Password Policy:

- Minimum 8 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one number

- At least one special character
- Cannot be same as last 3 passwords
- Password expiration: 180 days for students, 90 days for staff

Academic Data Security:

- Student Record Encryption: AES-256 for sensitive academic data
- Grade Information Protection: Privacy controls and access restrictions
- Assessment Data Security: Encrypted storage of exam and assignment records
- FERPA Compliance: Secure handling of educational records

Appendix E: Integration Specifications

Educational Platform Integration:

- LTI (Learning Tools Interoperability) support
- SCORM content package support
- Student Information System integration
- Library system connectivity

External Service Integration:

- Email notification services
- Video conferencing platforms (Zoom, Teams)
- Cloud storage services (Google Drive, OneDrive)
- Plagiarism detection services

Assessment Integration:

- Proctoring service integration
- Question bank providers
- Automated grading systems
- Accessibility testing tools

Appendix F: Business Rules and Constraints

Course Management Rules:

- Courses cannot be deleted after students enroll
- Course capacity cannot be reduced below current enrollment
- Instructors can only modify their own courses
- Course codes must be unique within the institution

Enrollment Rules:

- Students can enroll up to maximum course capacity
- Prerequisites must be completed before enrollment
- Enrollment periods are enforced by academic calendar
- Withdrawals have deadline restrictions

Grading Rules:

- Final grades cannot be changed after submission deadline
- All assignments must be graded before final grade calculation
- Grade modifications require administrator approval
- Academic integrity violations are flagged automatically

Appendix G: Future Enhancement Roadmap

Phase 2 Features:

- Mobile native applications for iOS and Android
- Advanced AI-powered learning recommendations
- Integration with virtual reality learning environments
- Blockchain-based credential verification
- Advanced proctoring and anti-cheating measures

Phase 3 Features:

- Artificial intelligence for personalized learning paths
- Advanced analytics with machine learning insights
- Integration with employer systems for workforce development
- Multi-language support for international programs
- Advanced accessibility features for diverse learners

Phase 4 Features:

- Virtual reality classroom experiences
- AI-powered teaching assistants
- Advanced competency-based learning management
- Integration with industry certification programs
- Global education network connectivity

Appendix H: API Summary

Course Management:

- POST /courses - Create new course
- GET /courses - Retrieve all available courses
- GET /courses/{id} - Retrieve specific course details
- PUT /courses/{id} - Update existing course
- DELETE /courses/{id} - Archive course

Enrollment Management:

- POST /enrollments - Enroll student in course
- GET /enrollments/student/{studentId} - Get student enrollments
- GET /enrollments/course/{courseId} - Get course enrollments
- PUT /enrollments/{id} - Update enrollment status

Assessment Management:

- POST /assessments - Create new assessment
- GET /assessments/course/{courseId} - Get course assessments
- POST /submissions - Submit assessment
- GET /submissions/{id} - Get submission details

User Management:

- POST /auth/login - User authentication
- POST /auth/logout - User logout
- GET /users/profile - Get user profile
- PUT /users/profile - Update user profile
- POST /auth/register - User registration

Appendix I: Component and Navigation Structure

React Components:

App.jsx:

- Main routing component with React Router
- Routes: "/" (Home), "/login" (Login), "/dashboard" (Dashboard), "/courses" (CourseList)
- Includes NavBar and authentication guards

NavBar.jsx:

- Application title: "LMS Lite - Learning Management System"

- Navigation links based on user role
- User profile dropdown with logout option

Dashboard.jsx:

- Role-based dashboard content
- Quick access to courses, assignments, grades
- Recent activity and notifications

CourseList.jsx:

- Display available courses
- Course search and filtering
- Enrollment capabilities for students

CourseDetail.jsx:

- Course information and content
- Module navigation and progress tracking
- Assignment and assessment access

Document Version Control

Version	Changes	Author
1.0	Initial SRS document creation for Learning Management System (LMS) Lite	Development Team